

0 8

A student investigated the effect of two antimicrobial substances, **J** and **K**, on the growth of *E. coli* bacteria.

She transferred *E. coli* cells using a sterilised pipette to make three identical cultures, **1**, **2**, and **3**. She then added:

- no antimicrobial substance to culture **1**
- antimicrobial substance **J** to culture **2**
- antimicrobial substance **K** to culture **3**.

She incubated the cultures for 24 hours, after which she determined the number of cells per mm³ in each culture.

0 8 . 1

The student used a sterilised pipette to transfer *E. coli* into each culture.

Suggest why the number of *E. coli* cells per mm³ in each culture after 24 hours might have been lower if the student had **not** used a sterilised pipette. Explain your answer.

[2 marks]

0 8 . 2

The student diluted 3 cm³ of culture **1** with 12 cm³ of water. She observed a sample of this diluted mixture using an optical microscope and counted 24 cells in 0.000 25 mm³ of the diluted mixture.

Use this information to calculate the number of cells per mm³ in **undiluted** culture **1**.

[2 marks]

Number of cells = _____ per mm³



0 8 . 3

After 24 hours, the student compared the number of cells per mm^3 in cultures **1**, **2** and **3**. She found:

- substance **J** killed 80% of the cells
- substance **J** killed twice as many cells as substance **K**.

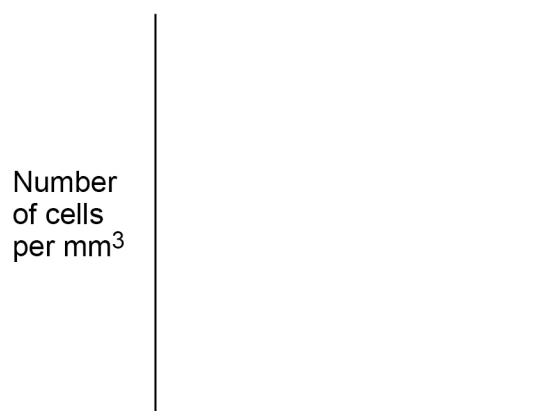
Using the axes shown in **Figure 9**, **sketch** a bar chart showing the results the student obtained from cultures **1**, **2** and **3**.

Do **not** draw a grid on the chart.

Do **not** include figures for the number of cells per mm^3

[3 marks]

Figure 9



7

Turn over for the next question

Turn over ►

